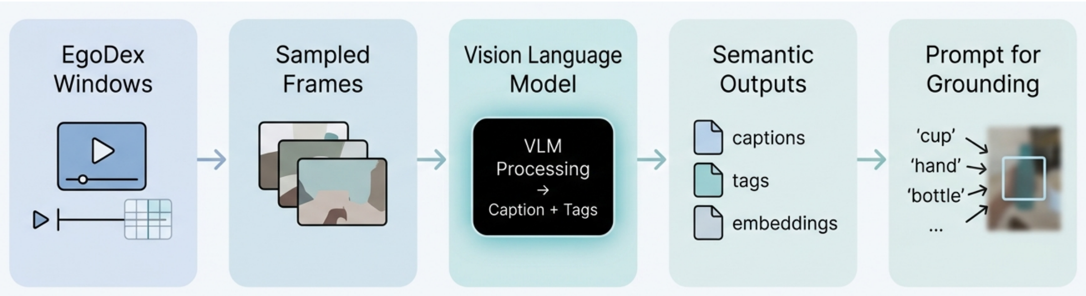
M.AI.ESTRO-Twin: Placement 1-Page Progress Report

Graham Pellegrini
0352804L
University of Malta
graham.pellegrini.22@um.edu.mt

I. Overview

This mid placement report summarises the current implementation milestone of the M.AI.ESTRO-Twin vision pipeline. My present placement contribution is a semantic aligner built over the open source EgoDex dataset and executed through a Docker based workflow on the University of Malta kr2280 shared GPU cluster. I currently convert indexed egocentric task windows into captions, structured tags, canonical semantic text, and embeddings for downstream policy and world model conditioning. This report outlines the current system structure, the semantic model comparison I have carried out so far, and the next planned extension into automated object grounding. I have not yet carried 6 DoF pose estimation forward in this checkpoint.

II. Implementation



I implemented this stage by building a window level semantic indexer around Hugging Face model assets. I loaded the required checkpoint weights, tokenizer, and processor files for each candidate generator, then paired them with SigLIP, a language image pre training model that maps text and images into a shared embedding space. This allowed me to assess whether the generated semantic descriptions stayed well aligned with the visual content of each EgoDex window. In the source code, I sampled representative frames, passed them through the semantic aligner to generate captions and structured tags, converted those tags into canonical semantic text, and wrote the aligned rows and manifest files needed for later reuse and multi GPU shard merging.

III. Results

The evaluated generator set includes Qwen3-VL-2B, Qwen3-VL-8B, LLaVA-v1.6-7B, Idefics3-8B, Qwen3.5-4B, Qwen3.5-9B, and SmolVLM2-2.2B. SigLIP was fixed as the operational embedding backend during the main semantic decision pass. The table below summarises the SigLIP 64 window bakeoff, where Quality is the semantic quality score and Sec./window is the runtime cost.

Model	Quality	Sec./window
Qwen3-VL-2B-Instruct	0.365	1.432
Qwen3-VL-8B-Instruct	0.431	2.435
LLaVA-v1.6-Mistral-7B	0.422	3.317
Idefics3-8B-Llama3	0.405	4.186
Qwen3.5-4B	0.434	5.763
Qwen3.5-9B	0.402	6.477
SmolVLM2-2.2B-Instruct	0.319	7.046

The measured trade off shows that larger models can improve raw quality, but the best operational balance for this placement stage remains Qwen3-VL-2B-Instruct because it is substantially faster while still producing stable semantic outputs. This operational choice was based on the full SigLIP bakeoff criterion, not only the two displayed metrics. The current result is therefore a working semantic preprocessing stage rather than a final manipulation model.

IV. Ongoing and Next Work

The next placement step is to extend the semantic stage into object grounding and binary mask generation using the produced object list as the prompt source for the bounding box unit. I intend this to automate the next module without waiting for manual user input. Reference [1] is relevant because it shows semantic descriptions being checked in an embedding space before being passed into a downstream module. Grounding DINO is then relevant as the next practical stage because it accepts text prompts and converts them into image space detections. Further work will determine whether an additional 6 DoF module is genuinely required for the downstream policy and world model setting or whether semantic and 2D grounding signals are sufficient.

References

[1] W. Zhang and J. Tao, "Automatic Semantic Alignment of Flow Pattern Representations for Exploration with Large Language Models," arXiv:2508.06300, 2025.

[2] T. Ren et al., "Grounding DINO 1.5: Advance the 'Edge' of Open-Set Object Detection," arXiv:2405.10300, 2024.